Ch 7.1-7.4: Polynomial regression, Basis functions, Cubic splines

Lecture 18 - CMSE 381

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Announcements

Last time:

- Lab - PCA/Ridge/Lasso

This lecture:

- 7.1 Polynomial regression
- 7.2 Step functions
- 7.3 Basis functions
- 7.4 Regression Splines

Announcements:

- Mon 3/20: Ch 7.1-7.4
- Weds 3/22: Ch 7.5, maybe 7.6?
- Friday lab
- Friday Homework #7 due

Section 1

Last time

High-Dimensional Data

data pts

variables

Low-Dimensions

$n \gg p$

- Low here means p is low, or at least small relative to n
- Can do all the stuff we've talked about so far

High-Dimensions

$n \ll p$

- Issues show up even if p is close to or slightly smaller than n
- Classical approaches not appropriate since lots of overfitting

What to do about it?

Be less flexible....

Key points

Ridge / Lasso

- regularization or shrinkage plays a key role in high-dimensional problems,
- appropriate tuning parameter selection is crucial for good predictive performance, and
- the test error tends to increase as the dimensionality of the problem increases, unless the additional features are truly associated with the response.

λ

Things get weird in high dimensions

- Curse of dimensionality
- Report results on an independent test set, or cross-validation errors.

More flexibility!!!

Section 2

Polynomial Regression

Polynomial regression

$p = 1$

Data :

x	y

Replace linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

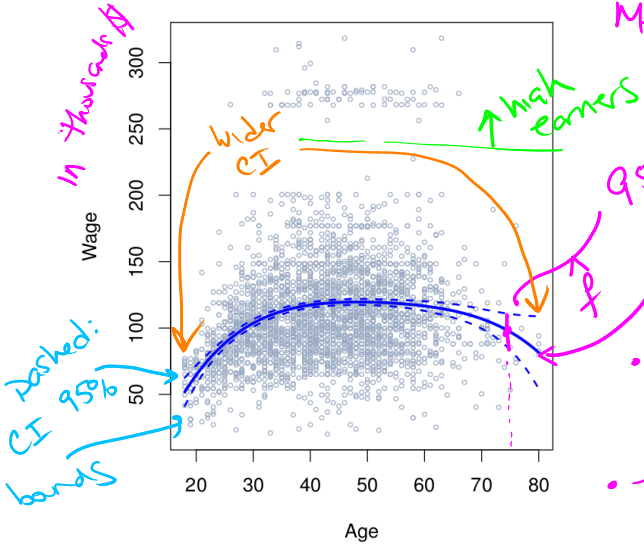
x^2 $\quad$ x^d

pass to linear regression

x	x^2	x^3	x^4	y

$d \leq 3$ or 4 $\rightsquigarrow$ overfitting

Example with wage data



Men in central Atlantic region of US

95% CI for age 75

$\hat{f}$

deg 4 polynomial

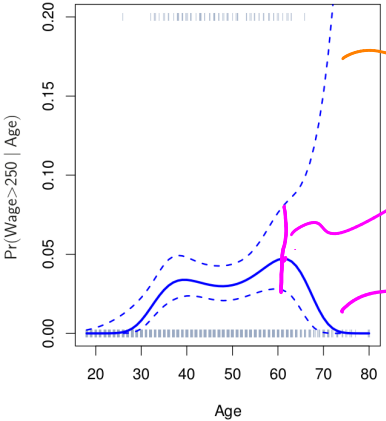
- Can still use variance calculations from lin reg
- That can be used to get pointwise variance $Var(\hat{f}(x_0))$

Classification version

logistic reg

polynomial

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



very wide on end pts

confidence interval @ age = 61

learned model

Section 3

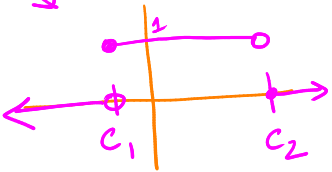
Step function

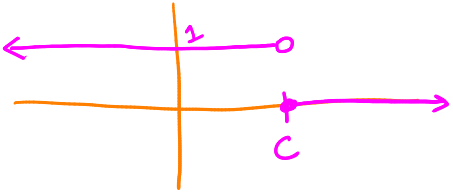
Step functions

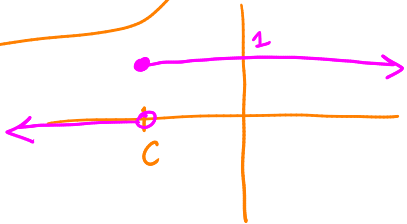
c is fixed # in $\mathbb{R}$

$\mathrm{I}(X < c)$ $\quad$ $\mathrm{I}(c_1 \le X < c_2)$ $\quad$ $\mathrm{I}(c \le X)$

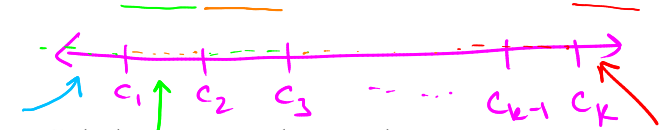
$$I(X<c) = \begin{cases} 1 & x < c \\ 0 & x \geq c \end{cases}$$







More on step function setup



$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$

These are sometimes called dummy variables

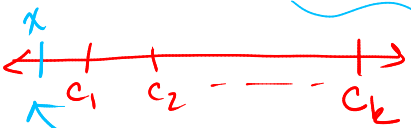
$$C_0(x) + C_1(x) + C_2(x) + \cdots + C_k(x) = 1$$

every x in exactly one interval

Step function: Learned model

No C_0 because its value can be determined by knowing $C_1(x), \ldots, C_K(x)$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

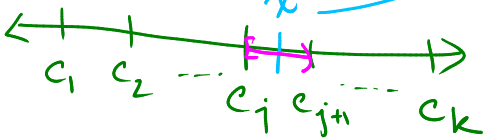


plug in this x $\rightsquigarrow$ $= \beta_0$ = mean value of y_i for $x_i \leq c_1$

Step function: Learned model

$$y_i = \underbrace{\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i)} + \varepsilon_i$$

x

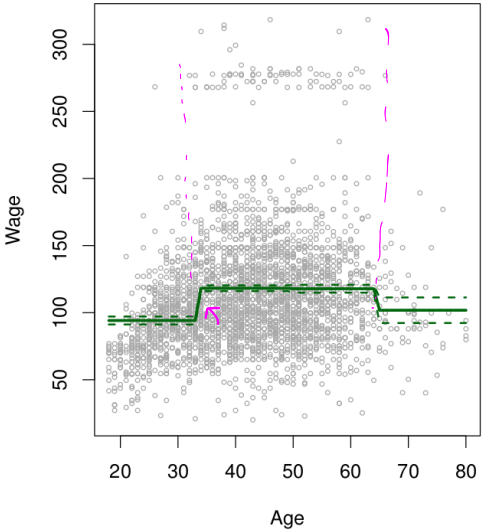


plug in:

$\beta_0 + \beta_j$

β_j is the avg increase in response for x in this interval rel to $x < c_1$

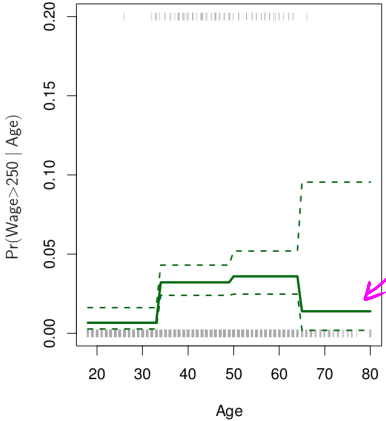
Step function example



- chose c_i's
  Maybe every 5 yrs?

Step function classification example

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



learned model

A few more comments on step functions

- Gives us a chance to break up the domain, avoid forcing global structure
  less "wiggly bits"
- Need to decide where the c_i's go
  Arbitrary unless there's natural split in x's

Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

Pick basis functions $b_1, \ldots, b_K$

Use least squares to find constants

Still use all inference tools from before

Prev: Polynomial $b_i(x) = x^i$

Step fcn: $b_i(x) = c_i(x)$

Fourier Series

Wavelets

Section 4

Regression Splines

Cubic splines

to do regression

Piecewise polynomials

Step functions = piecewise $d=0$ degree polynomials with knots @ c_i's

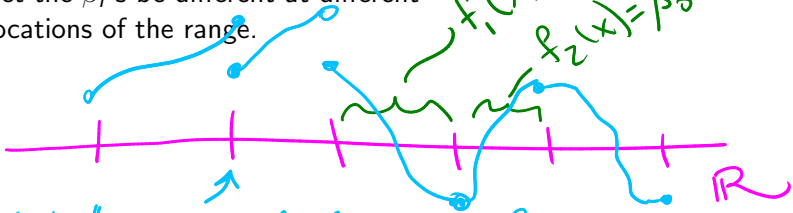
- Fit a polynomial regression

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

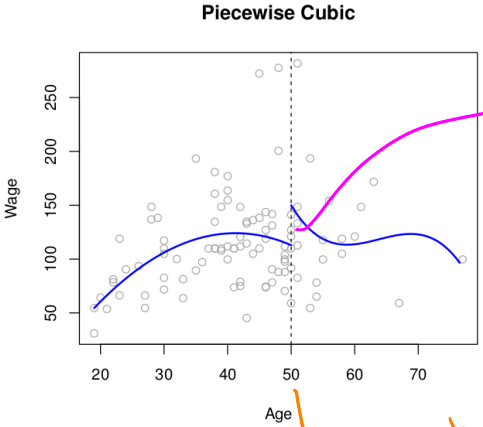
$f_1(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d$

$f_2(x) = \beta_0' + \beta_1' x + \beta_2' x^2 + \cdots + \beta_d' x^d$



"knots" = potential locations for discontinuities

Example of piecewise polynomial



yikes! discontinuity!

Example:

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

More knots = more flexibility

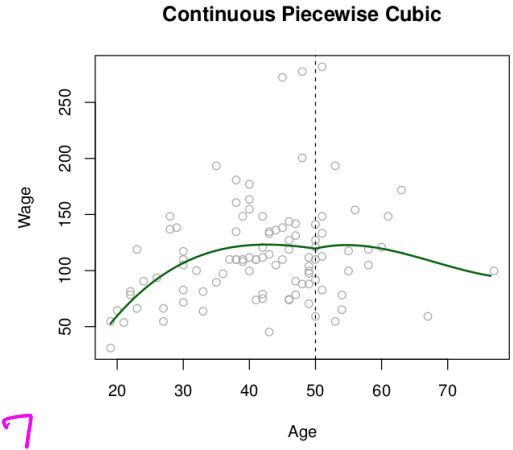
one knot @ age = 50

degrees of freedom here = 8

The fix

$d=3$

- Fit piecewise polynomial
- Require continuity at knots



In this example

deg freedom: $8-1=7$

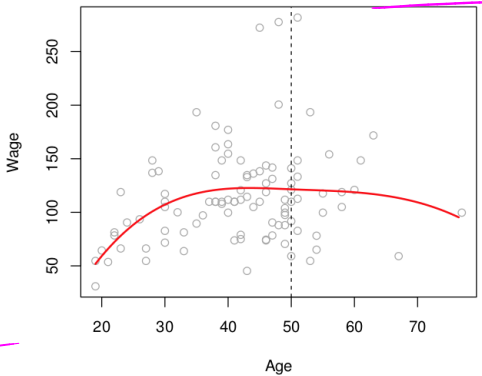
The better fix: Cubic splines

$\hookrightarrow d = 3$

cubic spline deg freedom w/ K knots $\boxed{4+K}$

- Fit piecewise polynomial
- Require continuity at knots
- Require the first and second derivatives to be continuous at knots

Results in "smooth" curve



$8 - 3$ constraints $= 5$ df

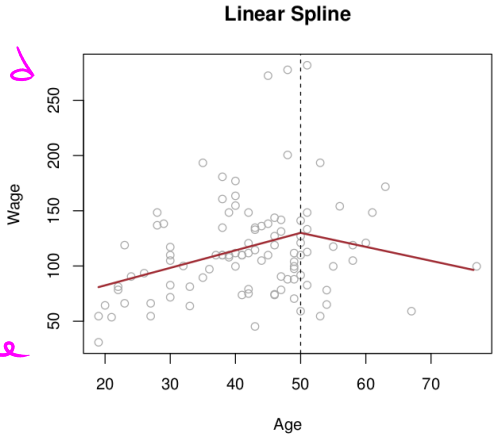
- f cont
- f' cont
- f'' cont

Generalized splines

- Can do for any degree d polynomial

$d=1$

- Require continuity up to the $(d-1)^{st}$ derivative



Spline basis representation

$df = K+4$

Want to pick b_i so that we represent a cubic spline with K knots as

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Truncated power basis function

z = location of knot

$z = \xi$ "x_i"

$$h(x, z) = (x - z)_+^3 = \begin{cases} (x - z)^3 & \text{if } x > z \\ 0 & \textit{else} \end{cases}$$

Desmos link: https://www.desmos.com/calculator/esucuulbgj

The basis for cubic splines

Given knots at $z_1, \cdots, z_K$

- X
- X^2
- X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$

$K+3$ functions

$K+4$ df w/ intercept

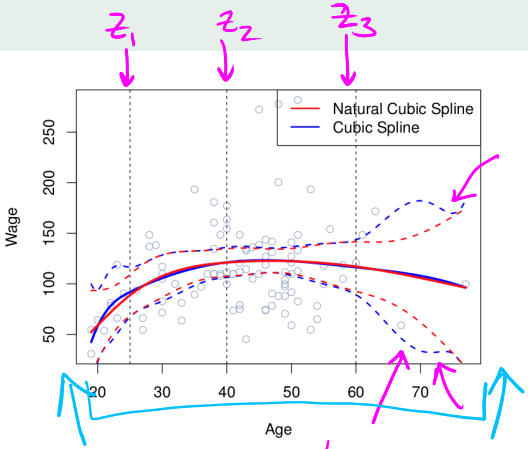
Notes on cubic splines

"Natural cubic spline"

- Regular cubic spline

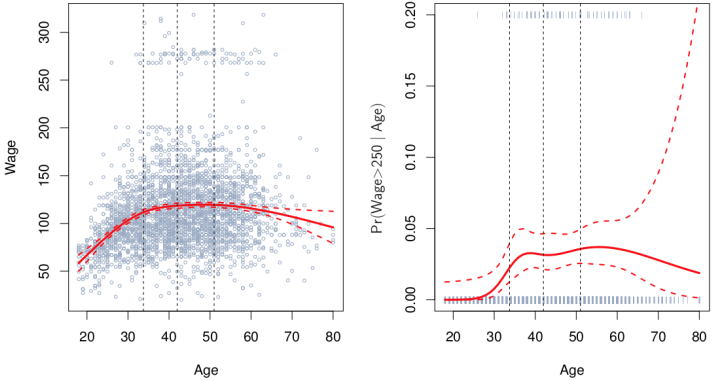
+

- boundary constraints forces model to be linear @ boundary



high variance at outer range of X

Where to put the knots?



Plan A: Uniform placement

Plan B: data driven approach

Here: knots @ data quantiles

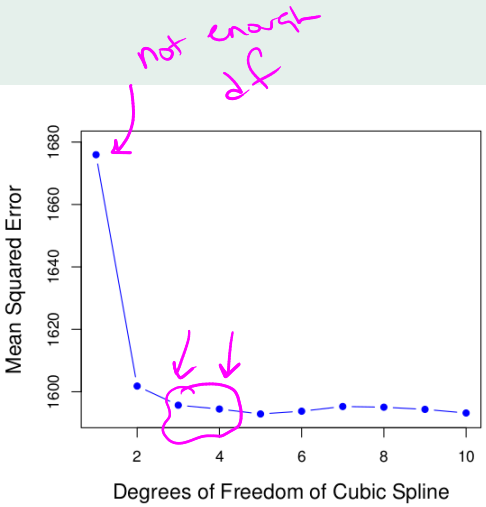
How many knots to use?

When in doubt, Cross-Validate.

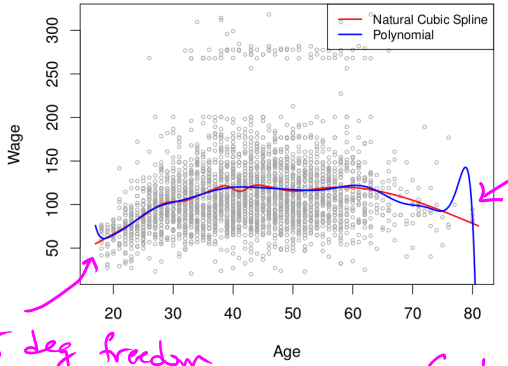
fix # knots

Do CV to get approx test error

Repeat



Cubic splines vs Polynomial Regression



ends tend to go wiggly for high degree polynomials

15 deg freedom cubic spline

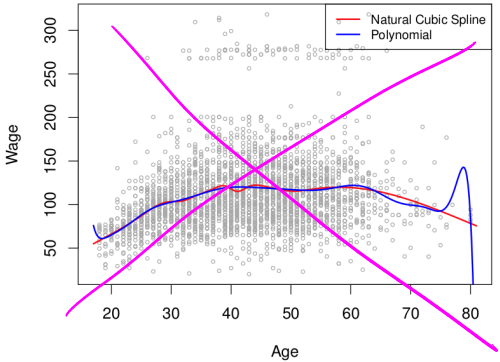
vs

15 deg polynomial

Cubic splines allow for flexibility using knots rather than having flexibility resulting in weird ends

~~Cubic splines vs Polynomial Regression~~

TL;DR



- Basis functions

$\beta_0 + \beta_1 b_1(x) + \dots + \beta_k b_k(x)$

- polynomials
- step fcns
- cubic splines
  - ↳ piecewise polynomials w/ smoothness constraints

Next time

Next time:

- 7.5, Maybe 7.7

This week:

- Mon 3/14: 7.1-7.4
- Weds 3/16: 7.5, maybe 7.7?
- Fri 3/18: Lab: 7.8
  - HW #7 Due Friday
  - Covers PCA/Ridge/Lasso Lab. Also content from today and Wednesday.

Almost exactly right timing. Watch out for a few duplicated slides that need to be fixed for next year (#13 maybe? And a TLDR was added)